Digital Library Perspectives

Sustainability of digital preservation at university libraries of Pakistan

Muhammad Hamza and Ahsan Ullah
Department of Library and Information Management, Superior University, Lahore, Pakistan

Received 12 August 2024
Revised 1 December 2024
3 March 2025
26 April 2025
21 June 2025
Accepted 21 June 2025

Abstract

Purpose – Digital preservation in university libraries is designed to foster long-term access to and stewardship of knowledge resources. This study aims to examine the sustainability of preservation policies, technologies and strategies.

Design/methodology/approach – The quantitative methodology was followed in this research by collecting data through questionnaires responded by Pakistani university librarians. Descriptive and inferential statistics were used to analyze the data.

Findings – The results reveal that sustainability measures for digital preservation policies and technologies are being taken at university libraries in Pakistan. Effective strategies have been adopted to create sustainability in the digital preservation of library material. An independent sample *t*-test showed no significant differences for gender or sector regarding the sustainability levels of policies, technologies or strategies. Those respondents who report low or no engagement in digital preservation activities ranked sustainability measures more favorably than those highly engaged in learning about digital preservation.

Originality/value – The current research outlines the dire necessity felt for feasible policies and practical strategies, supported by state-of-the-art technologies, to ensure perpetual access to hugely valued digital resources in university libraries in Pakistan.

Keywords Pakistan, Digitization, University libraries, Sustainability, Preservation, Policies

Paper type Research paper

1. Introduction

Technological advancements have made it much easier and faster to create, share and access information. Digital materials offer many advantages over traditional library resources, leading to a rapid increase in the creation, demand and use of digital content. Digital objects range from basic text files and photos to advanced Web-based content that fully uses technology. While technological innovations continue to advance, they also present some challenges (Ahmad *et al.*, 2023). Digital information resources are at risk from natural disasters, human errors, hardware failures, obsolescence and cyber threats, underscoring the need for robust preservation and security measures (Ahmad and Rafiq, 2024). These challenges may expose valuable information assets to obsolescence and loss. Digital preservation is crucial in this regard as it entails safeguarding enormous human knowledge for future generations (Njuguna and Kimote, 2024). Digital preservation fosters long-term access and stewardship of learning and knowledge resources. Patra and Sahoo (2022) noted that digital preservation is a crucial activity of modern library service, as it ensures the long-term access and use of digital resources.

Lack of institutional and national policies, unavailability of rich technological support and last but not least, the absence of appropriate measures may hamper sustainability of digital preservation (Ahmad and Rafiq, 2024; Ahmad *et al.*, 2023). Inadequate funding limits the procurement of advanced technologies and infrastructure necessary for digital preservation. As a result, libraries are not usually in a position to adhere to modern

Digital Library Perspectives
Vol. 41 No. 4, 2025
pp. 683-693
© Emerald Publishing Limited
2059-5816
DOI 10.1108/DLP-08-2024-0133

Downloaded from www.emerald.com/dlp/article-pdf/41/4/683/10047432/dlp-08-2024-0133en.pdf by Diponegoro University user on 10 November 2026

preservation techniques, which essentially increases the chances of data loss and the degradation of digital assets over time (Ahmad and Rafiq, 2024). The ineffective administrative support further intensifies these problems, as the staff within the libraries are not trained and equipped enough to develop policies and adopt effective digital preservation strategies.

Factors negatively affecting digital preservation may be removed by the adoption of sustainable digital preservation. "Sustainable digital preservation is about ensuring the long-term preservation of digital information while minimizing environmental, economic and social impacts." Traditional digital preservation focuses on technical challenges while sustainable digital preservation acknowledges resource consumption and ethical consideration (climate.sustainability-directory.com). Shimray and Ramaiah (2018) emphasized that the sustainability of digital preservation program requires robust measures like the adoption of standardized formats, data migration, technology watch and the use of redundant storage. Sustainability in digital preservation entails technical continuity, organizational resilience, financial durability or policy robustness. This sustainable program needs organizational framework, dedicated preservation teams, continuous staff training and its integration into institutional workflows, backed by long-term funding and policies. Diverse range of digital preservation strategies is available like replication, migration, normalization, emulation, selection of robust storage, distribution of backups, encapsulation, metadata attachment, online archiving, media refreshing and digital archaeology to meet challenges (Ahmad and Rafiq, 2023). Sustainability in digital preservation can be achieved through adoption of sustainable technologies, policies and strategies.

By considering the importance of sustainability of digital preservation, this study intends to examine the prevailing situation of sustainable digital preservation in university libraries of Pakistan. It tries to provide a deeper insight into the factors affecting the sustainability of digital preservation in these institutions. Assessing the attitudes of university librarians in Pakistan toward the sustainability of preservation policies, technologies and strategies will identify best practice and insights into factors influencing the sustainability of digital preservation. The findings of this research will help in giving insight to the regulatory bodies to work toward the development of institutional policies regarding digital preservation in Pakistan. It will help libraries in framing guidelines toward more effective and sustainable practices.

2. Objective

This research focuses on the opinions of academic librarians in Pakistan about sustainability in relation to preservation policies, technological infrastructure and the prevalent digital preservation strategies in the responding libraries. The outcomes will provide an insight to the regulatory bodies and university administration to design strategies for improving the implementation of best strategies in libraries. The following research objectives have been specified:

(1) to investigate the sustainability of digital preservation policies and program in the university libraries of Pakistan;

(2) to examine the sustainability of technology required for digital preservation program in these libraries; and

(3) to check the adoption of sustainable strategies among university librarians for digital preservation.

Downloaded from www.emerald.com/dlp/article-pdf/41/4/683/10047432/dlp-08-2024-0133en.pdf by Diponegoro University user on 10 November 2026

3. Literature review

3.1 Issues hampering digital preservation

Review of literature revealed different issues hampering the activities of digital preservation. Routhier Perry (2014) identified the technological obsolescence as core problem hindering the accessibility of digital objects. Routhier Perry (2014) postulated that one major challenge to digital preservation is the fact that it has a very fast rate of obsolescence. Digital formats and hardware have quick potential for becoming obsolete, and often, in the absence of access means, as a result of continuously evolving technology, digitized materials sustain losses; hence, this calls for preservation not only of content but also of access in changing times. The sustainability of digital preservation is hindered by issues like higher initial costs, lack of technological infrastructure, financial resources and skilled staff (Patra and Sahoo, 2022). Administrative, financial, technical and copyright issues impede the effective digital preservation of theses and dissertations at Central Library, Quaid-I-Azam University Islamabad (Arif *et al.*, 2024). Another challenge associated with digital preservation is security and privacy (Bellini and Tammaro, 2024).

3.2 Role of policies for sustainable digital preservation

Previous studies also examined the role of policies for sustainable digital preservation. A digital preservation policy needs to illustrate the fact that the institution in question is fully committed to the preservation of digital content for future use and that the necessary resources and support are available. Periodic reviews for such policies for digital preservation are also necessary, as the policies may need to accommodate new dimensions brought up by technology that continues to change at a fast rate (Sambo *et al.*, 2014). Ohio State University Libraries are a good example of the success to be achieved in the development and adoption of digital preservation policies. One-year process engaged a cross-functional team from the institutional repository, the information technology unit and special collections and archives. The resulting policy framework formalized the library's commitment to long-term stewardship, preservation and sustainable access to digital assets (Noonan, 2014). According to Shimray and Ramaiah (2018), comprehensive digital preservation policies must be set up that are concerned with and concentrated on the integrity of data, security and proper access control and be upgraded from time to time so that they keep pace with the fast-changing technologies. Masenya and Ngulube (2020) declared that policies provide the rules and procedures required to guarantee that digital assets are accessible and usable over time. The sustainability of digital preservation tactics in university libraries also depends on the establishment of strong policies and procedures. According to Patra *et al.* (2021), policies are required to maintain the accessibility and authenticity of digital collections over time in the sense that they provide a structured framework to address the variety of technical, organizational and economic challenges associated with digital preservation. Digital preservation policies provide the organizations with the complete course of action regarding the acquisition, ingestion, metadata, storage security measures, normalization, migration, media refreshing and encapsulation (Ahmad and Rafiq, 2022). Patra and Sahoo (2022) stressed that well-managed policies guide the organization to ensure the accessibility of digital materials.

3.3 Technology and sustainability of digital preservation

Adoption and preservation of innovative and modern technology has important role in digital preservation. The old technology, hardware and software used in creating and accessing digital information are preserved in their original form and environment. Another approach is metadata management, which is vital for tackling lineage in a digital object. Metadata will

Downloaded from www.emerald.com/dlp/article-pdf/41/4/683/10047432/dlp-08-2024-0133en.pdf by Diponegoro University user on 10 November 2026

be captured in an object to preserve the object and later be located and retrieved by the user. Digital archaeology is additional strategy for the rescue of content from damaged or decaying media in obsolete hardware and software environments. Digital archaeology can recover the digital materials from these extreme cases of risk and guarantee their long-term preservation. Migration can be regarded as a vital digital preservation technique for dealing with periodic copying, conversion and transfer of original information from one generation of technology to succeeding generations, aiming to keep digital information accessible and usable within a changing technological environment (Sambo *et al.*, 2014). According to Shimray and Ramaiah (2018), the sustainability of digital preservation strategies implies a multifaceted approach, requiring robust technological measures on standardized formats, data migration and redundant storage to avoid data loss and ensure interoperability. Mallery (2019) urged libraries to devote themselves to digital preservation technologies that show two principal characteristics: open standards and long-term format support. Open standards refer to the file formats and protocols that are publicly available and not controlled by a single vendor. This ensures a superior level of interoperability between different preservation systems. Standards such as these allow a certain level of flexibility in digital collections of libraries. Another important factor, according to Matlala (2019), is applying a well-thought-out data migration plan to systematically change storage media and formats, through which data is consistently moved to newer formats. Automated tools, in the form of preservation software for checking integrity and managing metadata, are important as well. These can increase the ease of preservation management, reduce the need for manual intervention and provide for greater authenticity of digital records. For the preservation of 3D objects in digital libraries, compatibility with future technologies and software is an issue that needs to be addressed (Guidi and Tammaro, 2024).

3.4 Strategies for achieving sustainable digital preservation

Various strategies and techniques have already been used by libraries to help achieve sustainable digital preservation. Digital preservation processes were adopted by Colorado State University Libraries through which university libraries preserve theses and dissertations. It successfully applied a digital archiving and preservation process that consists of identification, selection, packaging and archiving of local digital assets for the purposes of long-term access and migration (Oehlerts and Liu, 2013). Sambo *et al.* (2014) also considered professional development of staff as an effective way to inculcate awareness and commitment toward digital preservation among the staff. They also consider the application of digital archaeology, metadata management and migration as crucial strategies for sustainable digital preservation. Shimray and Ramaiah (2018) proposed an efficient organizational framework that include dedicated preservation teams, continuous staff training and its integration into institutional workflows, backed by long-term funding. Sharing resources and expertise between different institutions through interinstitutional collaboration enhances sustainability. Matlala (2019) considered the application of migration strategy, integrity checks, metadata attachment as essential strategies for ensuring the long-term access to digital objects.

Awamleh and Hamad's (2022) stressed investments in staff development for sustainability of digital preservation. Ahmad and Rafiq (2024) stressed that the librarians must have sufficient level of awareness about the digital preservation key components like key concepts, related challenges, the strategies to be implemented, the tools to be applied and the standards to be adhered for a long-term sustainable preservation program. Through working toward the removal of key barriers and the development and promotion of sustainable practices, libraries will be better able to ensure the long-term preservation of

Downloaded from www.emerald.com/dlp/article-pdf/41/4/683/10047432/dlp-08-2024-0133en.pdf by Diponegoro University user on 10 November 2026

digital collections (Ahmad and Rafiq, 2023; Ahmad *et al.*, 2023). Ahmad and Rafiq (2024) identified the strategies like replication, media migration, refreshing, normalization, emulation, encapsulation, online archiving development of trustworthy digital repositories, digital archives, distributed digital preservation networks and digital archaeology as the core strategies for sustainable preservation program. Bhat and Hasan (2024) advocated the community engagement and collaborations as crucial acts for sustainability of digital preservation. Awareness among librarians about institutional repository has positive and significant influence on sustainability (Esse and Haliso, 2024).

3.5 Gaps in literature about sustainability of digital preservation

Review of literature shows that a few studies have discussed sustainability in digital preservation but authors could not find any empirical study that collected data from libraries about the sustainability of digital preservation in university libraries. Paschalidou *et al.* (2022) also opined about lack of literature about sustainability in digital preservation while introducing the sustainability framework for digital preservation of cultural heritage. Dressler *et al.* (2024) also claimed that there is little literature about sustainable digital projects. Antoniazzi (2021) addressed the lack of literature on preservation of digital objects in film heritage institutions of Europe. The current study is designed to fill this gap by studying sustainability in digital preservation.

4. Research methodology

On the basis of nature of data, research approaches have been divided into four categories including quantitative, qualitative, mixed-methods and multimethods (Ullah and Ameen, 2022, 2023; Ullah *et al.*, 2024). The current study used quantitative approach to determine sustainability of digital preservation policies, technologies and strategies by university libraries.

A survey method was applied and a questionnaire was created by reviewing prior research and adjustments were made to align the measurements with the study's objectives. The questionnaire was divided into four parts. First part relates to demographic information. The subsequent three parts contained statements regarding the sustainability of policies, technologies and strategies used for digital preservation. Likert scale (Strongly Agree = 5, Agree = 4, Neutral = 3, Disagree = 2, Strongly Agree = 1) was used for measuring sustainability in policies, technology and strategies. The questionnaire had been tested to ensure its validity and reliability, for the study's objectives. For validity, questionnaire was discussed with three Pakistani researchers who have published articles in the field of digitization. For pilot testing, questionnaires were distributed to five library professionals working in the university libraries of Lahore. The recommendations of the respondents were incorporated in the questionnaire. To establish the tool's reliability, Cronbach's alpha coefficient was used across the entire questionnaire. The Cronbach's alpha value ranges between 0.85 and 0.972, indicating a high level of reliability.

According to Powell (2006), the population is made up of distinct things that share a characteristic. Population for the current study consists of librarians working in the central libraries of government and private section universities recognized by Higher Education Commission of Pakistan (HEC). There are 269 universities recognized by HEC in Pakistan (www.hec.gov.pk/english/universities/pages/recognised.aspx). Each university has established a central library to fulfill the needs of users.

Purposive and convenient sampling techniques were used to select sample. Purposive sampling was used to select librarians who were engaged in digitization activities. Convenient sampling was used to make a list of librarians whose contact details were

Downloaded from www.emerald.com/dlp/article-pdf/41/4/683/10047432/dlp-08-2024-0133en.pdf by Diponegoro University user on 10 November 2026

available on the Web page of each university library. Contact details of library professionals associated with digitization activities were also collected though personal phone calls.

For collection of data, questionnaires were sent to librarians performing digital preservation activities in central libraries. We were able to collect data from 103 university librarians but 2 questionnaires were discarded due to neutral response rate for policies, technologies and strategies of digital preservation. Data was collected for the completion of MPhil thesis of first author during March and April 2024. As part of the data collection procedure, the questionnaire was also created on Google Form. The link was sent to respondents through email and WhatsApp. Printed questionnaires were also sent to respondents working in university libraries of Lahore. For analysis of data, Statistical Package for the Social Sciences (SPSS), version 22, was used. Frequency distribution, mean and standard deviation were used to examine trends in sustainability of digital preservation. To determine the difference in opinions of respondents regarding sustainability of digital preservation on the basis of demographic data, the Independent Sample *t*-test was used.

5. Results

5.1 Demographic details

The demographic data described in Table 1 comprised details about the employment titles held by library staff, the nature of university, the respondents' gender and their awareness of the digitization of libraries. The library head (chief librarian) accounts for 11% of the total respondents. Nearly one-fourth of respondents (25.8%) were working as middle-level executives (deputy chief librarian, senior librarian and deputy librarian). Majority of participants (62.3%) were performing their duties as librarians or assistant librarians (initial level of recruitment).

Table 1. Demographic details of respondents

Demographic variables	No.	%
Designation		
Chief librarian	11	10.9
Deputy chief librarian	4	04.0
Deputy librarian	12	11.9
Senior librarian	11	10.9
Librarian/assistant librarians/others	63	62.3
Nature of library		
Government	69	68.3
Private	32	31.7
Gender of respondents		
Male	81	80.2
Female	20	19.8
Rate your awareness about library digitization		
High	51	50.5
Medium	07	6.9
Low	41	40.6
No	02	2.0
Total	*101*	*100.0*

Source(s): Authors' own work

Downloaded from www.emerald.com/dlp/article-pdf/41/4/683/10047432/dlp-08-2024-0133en.pdf by Diponegoro University user on 10 November 2026

One the basis of nature of universities, they were divided into government versus private universities. The 101 universities under study are divided into 31.7% private and 68.3% government sector. The third list shows that 80% of the 101 respondents were males. There were 20% female respondents. From the research, it was found that 50.5% had extensive knowledge in terms of digital library technology, whereas 40.6% were inexperienced, 6.9% had some knowledge and 2% had no knowledge at all on the same.

5.2 Sustainability of digital preservation policies

The mean score for digital preservation given in Table 2 indicated that libraries have adopted plans and policies for digital preservation of documents. Libraries have adopted disaster recovery plans to address loss of material and also communicate polices to stakeholders and users.

5.3 Sustainability of digital preservation technologies

The overall mean score is higher for sustainability of digital preservation technologies. The consideration of cutting-edge technologies in the near future to improve procedures for digital preservation with highest mean (4.02) showed a positive outlook. The mean scores for access through user-friendly interfaces ($M = 3.92$), use of standardized file formats ($M = 3.89$) and availability of dedicated personnel have expertise of digital technology show the higher level of sustainability of digital preservation technologies (see Table 3).

5.4 Sustainability of digital preservation strategies

Mean values given in Table 4 show that university libraries are making efforts to create sustainability in digital preservation strategies. The highest average score is for regularly backs up to its digital resources to maintain data integrity and avoid data loss. Different strategies such as the security of digital assets ($M = 3.84$), checking digital materials for any signs of degradation or technical problems ($M = 3.80$), to identify and solve dangers related to digital preservation ($M = 3.80$) and creation of metadata have been used frequently. Collaboration with other institutions for digital preservation ($M = 3.77$) has also been used for sustainability of digital preservation strategies.

5.5 Influences of sector, gender and level of awareness on sustainability of digital preservation

In case of results of a comparison of university libraries, both government and private, with regard to how long term their digital preservation strategies, technology and policies will last. The mean score for private university libraries is slightly higher than that for

Table 2. Sustainability of digital preservation policies

Sustainability of policies	Mean	SD
Library has future plans to improve and expand digital preservation efforts	4.09	0.87
Library has a digital preservation policy, which includes guidelines for selecting materials for preservation	3.85	0.99
Library has a well-documented disaster recovery plan to address potential risks to digital materials	3.81	1.05
Library communicates its digital preservation practices and policies to stakeholders and users	3.76	0.98

Source(s): Authors' own work

Downloaded from www.emerald.com/dlp/article-pdf/41/4/683/10047432/dlp-08-2024-0133en.pdf by Diponegoro University user on 10 November 2026

Table 3. Sustainability of digital preservation technologies

Sustainable technologies	Mean	SD
Library considers the adoption of emerging technologies to enhance digital preservation practices in the near future	4.02	0.84
Library ensures user access to preserved digital materials through user-friendly platforms or interfaces	3.92	0.98
Library uses standardized file formats for long-term preservation of digital content	3.89	0.87
Library has dedicated personnel with expertise in digital technology	3.79	1.03

Source(s): Authors' own work

Table 4. Sustainability of digital preservation strategies

Sustainable strategies	Mean	SD
Library regularly backup's digital materials to ensure data integrity and protection against loss	3.94	0.94
Library uses redundant storage solutions to enhance the security of digital assets	3.81	0.92
Library regularly monitors digital materials for any signs of degradation or technical issues	3.80	0.90
Library regularly assesses risk to identify and mitigate the associated threats for digital preservation	3.80	1.03
Library's digital preservation process uses appropriate metadata creation for enhanced discoverability	3.78	0.92
Library collaborates with other institutions or experts to enhance digital preservation efforts	3.77	0.98

Source(s): Authors' own work

government sector university libraries in the sustainability of digital preservation policies, technologies and strategies. Both types of libraries have a similar level of sustainability in their digital preservation strategies.

Table 5 reports the findings of a comparison on the basis of gender in the sustainability of policy, technologies and methods for the preservation of digital objects. For female respondents, the mean score for sustainability of digital preservation is higher than male but significant differences do not exist.

Independent Sample *t*-test was used for examining difference of opinion toward the sustainability of digital preservation policies, technologies and strategies between respondents having a high or medium level of awareness (high/medium) and those having a low or no knowledge (low/none). The results show that respondents who have higher knowledge of digitization consider that university libraries are making fewer efforts for sustainability of digitization.

6. Discussion

The results reveal that respondents perceive that digital preservation policies, technologies and strategies are being taken at university libraries in Pakistan. The higher values of means for policies show that university library have developed effective policies for digital

Downloaded from www.emerald.com/dlp/article-pdf/41/4/683/10047432/dlp-08-2024-0133en.pdf by Diponegoro University user on 10 November 2026

Table 5. Influences of sector, gender and level of awareness on sustainability of digital preservation

Sr. no.	Sector of universities	Means Public (69)	Private (32)	*t*-value	Sig.(two-tail)
1	Digital preservation policies	3.81	3.97	−1.02	0.31
2	Digital preservation technologies	3.89	3.92	−0.20	0.84
3	Digital preservation strategies	3.81	3.82	−0.04	0.97
	Gender influences	*Male (81)*	*Female (20)*	*t-value*	*Sig.(two-tail)*
1	Digital preservation policies	3.82	4.02	−1.29	0.21
2	Digital preservation technologies	3.87	4.05	−1.12	0.27
3	Digital preservation strategies	3.79	3.92	−0.81	0.42
	Level of awareness of respondents about digital preservation	*High/medium (58)*	*Low/none (43)*	*t-value*	*Sig.(two-tail)*
1	Digital preservation policies	3.49	4.36	−7.36	0.00
2	Digital preservation technologies	3.55	4.38	−7.23	0.00
3	Digital preservation strategies	3.45	4.30	−6.75	0.00

Source(s): Authors' own work

preservation of resources. Such policies must continue to overcome the technical, organizational and economic problems of digital preservation so that digital collections remain accessible, authentic and serviceable for a long time (Patra *et al.*, 2021). The development of evidence-based policies is necessary to forestall the loss, degradation or obsolescence of any digital collection without risk, hence securing investments in education and culture.

Digital preservation technologies are greatly concerned with sustainability for the sake of maintaining digital collections. The university libraries are implementing technologies to uphold open standards, formats that can be supported in the long term and interoperability between preservation systems that facilitate the migration of digital content into new systems with ease. Libraries need to have data migration plans together with automated integrity checking tools, metadata management and other measures to reduce manual interventions and enhance the authenticity of digital records (Mallery, 2019).

Effective strategies have also been adopted to create sustainability in digital preservation of library material. Adequate digital preservation strategies have been implemented in university libraries. Nevertheless, in the face of rapid technological obsolescence, access to digitized materials requires continued migration (Routhier Perry, 2014). It is recommended that library associations in Pakistan (Pakistan Library Association/Pakistan Library Club) should conduct workshops and seminars to create awareness about digital preservation. Possibility of collaboration may also be explored by working with other research support stakeholders such as Information Technology Department and Research Office within the university to enhance the quality of digital preservation.

The values of t for Independent Sample *t*-test show the variance of opinion among respondents on the basis of demographic variables. Independent Sample *t*-test showed no significant differences for gender. It shows similarity of views between male and female. There are no differences regarding the sustainability levels of policies, technologies or strategies for government and private sector libraries. Those respondents who report low or no engagement in digital preservation activities ranked sustainability measures more favorably than those highly engaged in learning of digital preservation. The more informed

Downloaded from www.emerald.com/dlp/article-pdf/41/4/683/10047432/dlp-08-2024-0133en.pdf by Diponegoro University user on 10 November 2026

respondents were of the view that digital preservation policies, technologies and strategies are not enough for streamlining digital preservation.

7. Conclusion

Digital preservation in university libraries is quite important for the long-term access and stewardship system connected with knowledge resources. The current research identified the level of sustainability of the digital preservation policies, technologies and strategies at university libraries in Pakistan. The current research has implications for research and practices. This can enhance trustworthiness of digital content for long-term accessibility and availability in university libraries in Pakistan. The research results would, therefore, also greatly contribute to the growth of institutional policies on digital preservation in Pakistan and support libraries in the establishment of guidelines concerning more factual and sustainable practices. The research will also be very important in designing knowledge-based systems that are capable of being modified and evolving by adjustment to meet the changing information and knowledge necessities of the library's users. By breaking some of the tidal waves – barriers – and encouraging the greatest and leading practices, university libraries in Pakistan can be certain to cope with or manage the preservation and accessibility of their digital collections in the long term, able to safeguard the nation's academic and cultural heritage. The results inform strategies for improving digital library services in developing nations. The sustainability of the approaches that are adopted in university libraries to ensure digital preservation is a necessity in any information society. Advanced technological solutions, sustainable organizational models and allied activities are necessary to achieve sustainability in digital projects. University administrative should support in sustaining digital collections and university librarians' skills for effective digital preservation.

References

Ahmad, R. and Rafiq, M. (2022), "Assessing the preparedness of university libraries for digital preservation", *The Journal of Academic Librarianship*, Vol. 48 No. 6, p. 102617.

Ahmad, R. and Rafiq, M. (2023), "Global perspective on digital preservation policy: a systematic review", *Journal of Librarianship and Information Science*, Vol. 55 No. 3, pp. 859-867.

Ahmad, R. and Rafiq, M. (2024), "Digital preservation practices for information resources in university libraries of Pakistan", *Online Information Review*, Vol. 48 No. 4, pp. 725-745.

Ahmad, R., Rafiq, M. and Arif, M. (2023), "Barriers to preservation for digital information resources in university libraries of Pakistan", *Restaurator. International Journal for the Preservation of Library and Archival Material*, Vol. 44 No. 3, pp. 221-243.

Antoniazzi, L. (2021), "Digital preservation and the sustainability of film heritage", *Information, Communication and Society*, Vol. 24 No. 11, pp. 1658-1673.

Arif, M., Mirza, K.B. and Hamid, M. (2024), "Investigating digitization and digital preservation strategies for theses and dissertations: a case study of Quaid-I-Azam university, Pakistan", *Global Knowledge, Memory and Communication*, doi: 10.1108/GKMC-11-2023-0447.

Awamleh, M.A. and Hamad, F. (2022), "Digital preservation of information sources at academic libraries in Jordan: an employee's perspective", *Library Management*, Vol. 43 Nos 1-2, pp. 172-191.

Bellini, E. and Tammaro, A.M. (2024), "Cybersecurity for digital libraries: an interview with Emanuele Bellini", *Digital Library Perspectives*, Vol. 40 No. 2, pp. 348-355.

Bhat, J.A. and Hasan, S. (2024), "A comprehensive overview of digital preservation in the digital library landscape", *Academic Libraries*, Vol. 220.

Downloaded from www.emerald.com/dlp/article-pdf/41/4/683/10047432/dlp-08-2024-0133en.pdf by Diponegoro University user on 10 November 2026

Dressler, V.A., Sweetser, M., Pavlik, N., Hawkins, M.C. and Harrington, K. (2024), "Beyond the grant: cultivating sustainable next steps", *Journal of Contemporary Archival Studies*, Vol. 11 No. 1, p. 5.

Esse, U. and Haliso, Y. (2024), "Facilitating conditions and institutional repository sustainability by librarians in public university libraries in Nigeria", *Digital Library Perspectives*, Vol. 40 No. 1, pp. 4-22.

Guidi, G. and Tammaro, A.M. (2024), "Embracing 3D and immersive technologies in digital libraries: interview with Gabriele Guidi", *Digital Library Perspectives*, Vol. 40 No. 1, pp. 148-153.

Mallery, M. (2019), "Digital preservation in libraries: preparing for a sustainable future".

Masenya, T.M. and Ngulube, P. (2020), "Factors that influence digital preservation sustainability in academic libraries in South Africa", *South African Journal of Libraries and Information Science*, Vol. 86 No. 1, pp. 52-63.

Matlala, E. (2019), "Long-term preservation of digital records at the university of KwaZulu-Natal archives", *Journal of the South African Society of Archivists*, Vol. 52, pp. 95-109.

Njuguna, A. and Kimote, Z. (2024), "Preserving the past, ensuring the future: challenges and innovations in digital preservation of information", 7th Annual International Conference 2024.

Noonan, D. (2014), "Digital preservation policy framework: a case study".

Oehlerts, B. and Liu, S. (2013), "Digital preservation strategies at Colorado State University libraries", *Library Management*, Vol. 34 Nos 1-2, pp. 83-95.

Paschalidou, E., Fafet, C. and Milios, L. (2022), "A strong sustainability framework for digital preservation of cultural heritage: introducing the eco-sufficiency perspective", *Heritage*, Vol. 5 No. 2, pp. 1066-1088.

Patra, S. and Sahoo, J. (2022), "A literature review on digitization in libraries and digital libraries", *Preservation, Digital Technology and Culture*, Vol. 51 No. 1, pp. 17-26.

Patra, S., Sahoo, J.D. and Mohanty, B. (2021), "Research on digital preservation: an empirical analysis", *Library Philosophy and Practice*, pp. 1-16.

Routhier Perry, S. (2014), "Digitization and digital preservation: a review of the literature", *School of Information Student Research Journal*, Vol. 4 No. 1, p. 4.

Sambo, A.S., Omeluzor, S.U. and Usman, A.S. (2014), "Awareness of digital preservation strategies by librarians in Nigeria", *International Journal of Information Research*, Vol. 3 No. 4.

Shimray, S.R. and Ramaiah, C.K. (2018), "Digital preservation strategies: an overview", 11th National Conference on Recent Advances in Information Technology-2018, pp. 46-54.

Ullah, A. and Ameen, K. (2022), "Examining the use of methodological pluralism in library and information science empirical research produced by Pakistani authors", *Malaysian Journal of Library and Information Science*, Vol. 27 No. 2, pp. 97-113.

Ullah, A. and Ameen, K. (2023), "Growth, subject areas, and application of research methods in user studies: a content analysis of articles produced by Pakistani authors", *Journal of Librarianship and Information Science*, Vol. 55 No. 4, pp. 1068-1078.

Ullah, A., Hayat, S. and Ameen, K. (2024), "Impact of research characteristics and methodologies used In LIS articles on citation count", *Journal of Librarianship and Information Science*, doi: 10.1177/09610006241269139.

Corresponding author
Ahsan Ullah can be contacted at: ahsanullah_libr@yahoo.com

For instructions on how to order reprints of this article, please visit our website:
www.emeraldgrouppublishing.com/licensing/reprints.htm
Or contact us for further details: **permissions@emeraldinsight.com**

Downloaded from www.emerald.com/dlp/article-pdf/41/4/683/10047432/dlp-08-2024-0133en.pdf by Diponegoro University user on 10 November 2026